

Propensity scores approaches with R

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1 Summary of model and weights

Logit model

```
psreg <- glm(treat ~ age + educ + CoVar1 ... + Covar10, data = lalonde, family = binomial(link = 'logit'))
```

Make unbalanced weights

```
lalonde <- lalonde %>%  
  mutate(prob = predict(psreg, type = 'response')) %>%  
  mutate(invwt = treat/prob + (1-treat)/(1-prob))
```

2 Balance: Standardized differences (SMD or STD)

2.1 Stddiff

Only for unadjusted baseline table, not for IPTW (It cannot use weights)

<https://cran.r-project.org/web/packages/stddiff/stddiff.pdf>

2.2 PSW

<https://cran.r-project.org/web/packages/PSW/PSW.pdf>

psw.balance: Balance checking using standardized mean difference

2.3 TableOne (recommended)

2.3.1 SMD (unweighted)

<https://cran.r-project.org/web/packages/tableone/vignettes/smd.html>

tableone package itself

```
library(tableone)
```

```
tabUnmatched <- CreateTableOne(vars = vars, strata = " treat", data = baseline, test = FALSE)
```

```
## Show table with SMD
```

```
print(tabUnmatched, smd = TRUE)
```

2.3.2 SMD (weighted)

<https://rpubs.com/sar03/642530>

Logistic regression to predict treat (numeric values 0 or 1)

```
mi <- glm(treat ~ CoVar1 + ... CoVar2, data=baseline, family = binomial(link = 'logit'))
```

Weights

```
#p1 proportion of patients in treat==1
```

```
#p0 proportion of patients in treat==0
```

```
p0<-as.numeric(summary(baseline$treat)[1])/nrow(baseline))
```

```
p1<-as.numeric(summary(baseline$treat)[2])/nrow(baseline))
```

```
# weights
```

```
baseline <- baseline %>%
```

```
  mutate(prob_i = predict(mi, type = 'response')) %>%
```

```
  mutate(IPTW_i = p0*treat/prob_i + p1 *(1-treat)/(1-prob_i))
```

Weighted data

```
DataIptw_i <- svydesign(ids = ~ 1, data = baseline, weights = ~ baseline$IPTW_i)
```

Table with weighted data

```
## Construct a table (This is a bit slow.)
```

```
# De vegades a vars cal passar-li els noms per a que funcioni !
```

```
# Ex1: colnames(baseline[-c(24,25)])
```

```
# Ex2: variables<-baseline[,c(3:23)] ; vars=names(variables[,])
```

```
# els números als que es fa referència als exemples, cal adequar-los
```

```
tabWeighted_i <- svyCreateTableOne(vars = ListOfvars, strata = "treat", data = DataIptw_i)
```

```
print(TabIPTW1, smd = TRUE)
```

Count covariates with important imbalance

```
addmargins(table(ExtractSmd(tabUnmatched) > 0.1))
```

3 IPTW

Fit model

```
psModel1 <- glm(formula = treat ~ [covariates with bad SMD], family = binomial(link = "logit"), data = datos)
```

#You need to find the best combination of covariate variables in the model to get best SMD combination

....

3.1 IPW

Hi ha multitud d'opcions a R, però millor useu la sintaxi de TableOne

https://statsnotebook.io/blog/analysis/marginal_structural_model_ipw/
library(ipw)

```
weight <- ipwpoint(exposure = can_1, family = "binomial", link = "logit",  
  numerator = ~ 1,  
  denominator = ~ failed_0+peer_can_0+antisocial_0+family_drug_use+sex+can_0,  
  trunc = 0.01, data = as.data.frame(currentDataset))
```

ipwpoint, for estimating inverse probability weights in a point treatment situation.
ipwplot, to plot the distribution of inverse probability weights.

3.2 Others

Twang (2021)

<https://cran.r-project.org/web/packages/twang/vignettes/ipw.pdf>

PSweight (2021)

<https://arxiv.org/pdf/2010.08893.pdf>

Table 1: Comparisons of existing R packages that implement propensity score weighting with discrete treatments. Binary treatments and additive estimands are implemented in all packages, and therefore those two columns are omitted.

	Multiple treatments	Balance diagnostics	IPW/ATT weights	OW/other weights	Ratio estimands	Augmented weighting	Nuisance-adj variance	Optimal trimming
PSweight	✓	✓	✓	✓	✓	✓	✓	✓
twang	✓	✓	✓	×	×	×	×	×
CBPS	✓	✓	✓	×	×	✓	✓	×
PSW	×	✓	✓	✓	✓	✓	✓	×
optweight	✓	×	✓	×	×	×	×	×
ATE	✓	✓	✓	×	×	×	✓	×
WeightIt	✓	×	✓	✓	×	×	×	×
causalweight	✓	×	✓	×	×	✓	×	×
sbw	×	✓	✓	×	×	×	×	×

✓ indicates that the functionality is currently implemented in the package; × indicates otherwise.

References: **twang** (Version 1.6): Ridgeway et al. (2020); **CBPS** (Version 0.21): Fong et al. (2019); **PSW** (Version 1.1-3): Mao and Li (2018); **optweight** (Version 0.2.5): Greifer (2019); **ATE** (Version 0.2.0): Haris and Chan (2015); **WeightIt** (Version 0.10.2): Greifer (2020); **causalweight** (Version 0.2.1): Bodory and Huber (2020); **sbw** (Version 1.1.1): Zubizarreta and Li (2020).

cobalt Package (2022)

Covariate Balance Tables and Plots

<https://ngreifer.github.io/cobalt/articles/cobalt.html>

love.plot

https://cran.rstudio.com/web/packages/cobalt/vignettes/cobalt_A4_love.plot.html

3.3 Examples

- <https://www.r-bloggers.com/2020/12/inverse-probability-treatment-weighting/> (recommended)
- http://rogosateaching.com/somgen290/cc_5.pdf
- <https://www.biostat.jhsph.edu/~estuart/HarderStuartAnthony09SampleRCode.R>
- <https://rpubs.com/sar03/642530>

4 Matching

<https://cran.r-project.org/web/packages/MatchIt/vignettes/matching-methods.html>

```
## PS matching  
library(Matching)
```

```
MATCH  
cem, http://gking.harvard.edu/cem/  
Matching, http://sekhon.berkeley.edu/matching  
MatchIt, http://gking.harvard.edu/matchit  
optmatch, http://cran.r-project.org/web/packages/optmatch/index.html  
PSAgraphics, http://cran.r-project.org/web/packages/PSAgraphics/index.html  
rbounds, http://cran.r-project.org/web/packages/rbounds/index.html  
twang, http://cran.r-project.org/web/packages/twang/index.html
```

5 Bibliography

5.1 Publicació

ipw: An R Package for Inverse Probability Weighting

<https://www.google.com/url?sa=t&rct=j&q=&esrc=s&source=web&cd=&cad=rja&uact=8&ved=2ahUKEwikIPuV29HpAhVB8OAKHRgyBxgQFjABegQIBBAB&url=https%3A%2F%2Fwww.jstatsoft.org%2Farticle%2Fview%2Fv043i13%2Fv43i13.pdf&usq=AOvVaw2NF7nRF1fodRpCAG621OnD>

5.2 Llibre en R

<https://books.google.es/books?id=4wecDQAAQBAJ&printsec=frontcover&dq=practical+propensity+score+methods+using+R&hl=es&sa=X#v=onepage&q=practical%20propensity%20score%20methods%20using%20R&f=false>

6 Tutorials

6.1 Video complet tutorial

MOLT Recomenable!!!

<https://www.coursera.org/lecture/crash-course-in-causality/data-example-in-r-le48W>

6.2 Web

Matching Weights Tutorial. Kazuki Yoshida. 2016-07-26

http://rstudio-pubs-static.s3.amazonaws.com/198545_5fae2128569a4e4a80627bc87c71d1e2.html

```
tab1Unadj <- CreateTableOne(vars = covariates, strata = "treat", data = tutoring)
```

```
tutoringSvy <- svydesign(ids = ~ 1, data = tutoring, weights = ~ mw)
```

```
tab1Mw <- svyCreateTableOne(vars = covariates, strata = "treat", data = tutoringSvy)
```

7 Analyses using PS

7.1 T-test

wtd.t.test

<https://rdr.io/cran/weights/man/wtd.t.test.html>

7.2 Weighted mean

weighted.mean

<https://www.rdocumentation.org/packages/stats/versions/3.6.2/topics/weighted.mean>

7.3 Weigthed %

<https://rdr.io/cran/weights/man/wpct.html>

7.4 Weighted logistic

`glm(Outcome ~ treat, weights = wt, data = data, family = binomial("logit"))`

7.5 Survival

7.5.1 Kaplan-Meier

Kaplan-Meier

<http://www.sthda.com/english/wiki/survival-analysis-basics>

7.5.2 Cox And weighted Cox

Cox

<http://www.sthda.com/english/wiki/cox-proportional-hazards-model>

<https://cran.r-project.org/web/packages/survival/survival.pdf>

```
Coxph(formula, data = , weights, subset,
      na.action, init, control,
      ties = c("efron", "breslow", "exact"),
      singular.ok = TRUE, robust,
      model = FALSE, x = FALSE, y = TRUE, tt, method = ties,
      id, cluster, istate, statedata, nocenter = c(-1, 0, 1), ...)
```